

Project: Firearm Information Repository System (FIRS)

1(a) Actors, Use Cases, and Relationships

Primary Actors

- Customers
- Authorized Dealers
- Government Authority
- Authorized Users
- System Admin

Use Cases

- Register Account
- Login
- Apply for Firearm License
- Renew Firearm License
- Check License Status
- Browse Gun Catalog
- View Gun Details
- Request Dealer Suggestion
- Apply for Selling Permit
- Renew Selling Permit
- Check Selling Permit Status
- Review License Application
- Approve License Application
- Reject License Application
- Review Selling Permit
- Approve Permit
- Reject Permit

- Update Firearm Sales Information
- Transfer Firearm Ownership
- Search Firearm's Criminal Activity
- Manage Firearm Inventory
- Monitor Welfare Usage
- Manage User Accounts
- Manage Firearm Database
- Manage Dealer Records
- Maintain System Database

Key Relationships

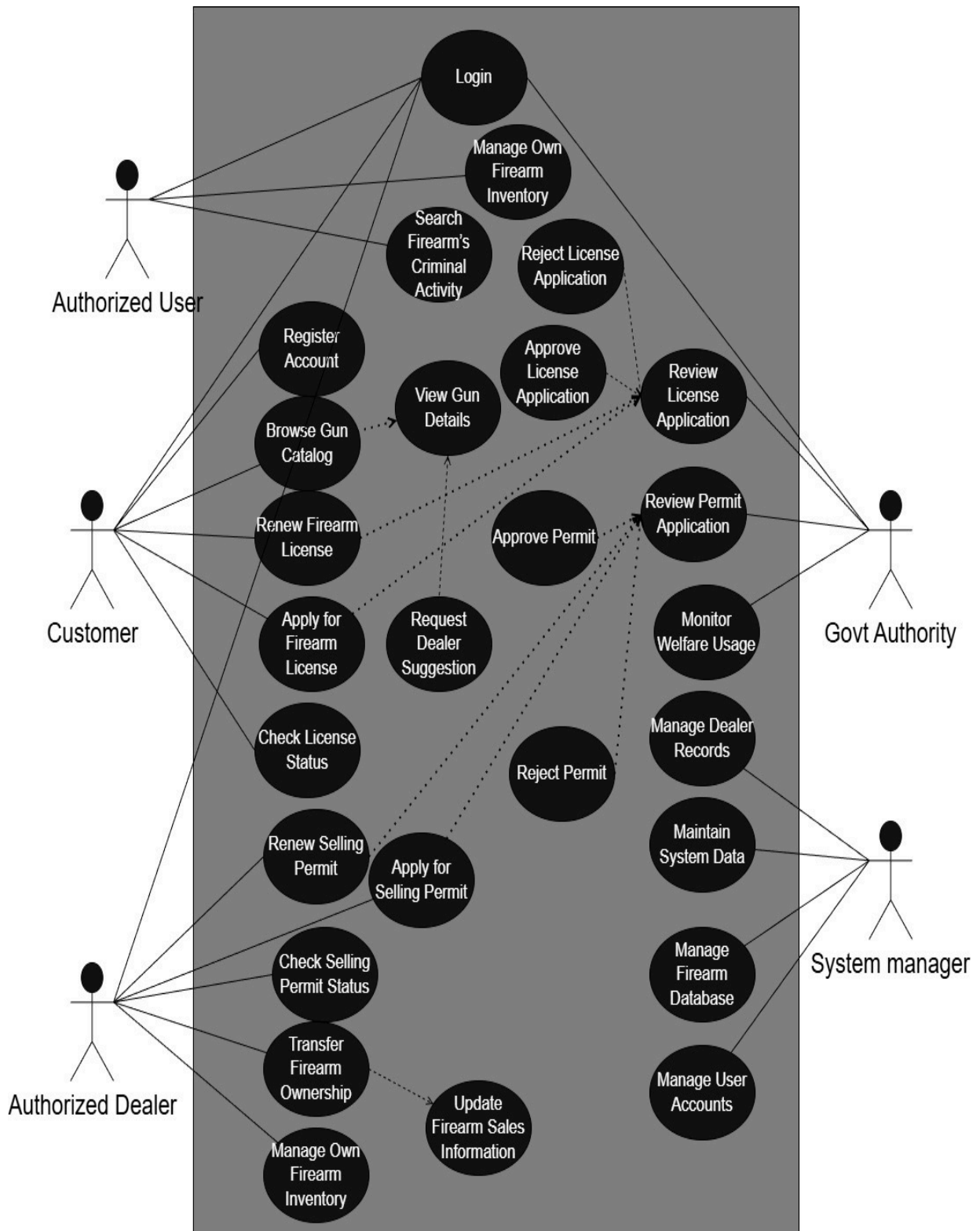
Include «include» Relationships

- Apply for Firearm License «include» Review License Application
- Renew Firearm License «include» Review License Application
- Apply for Selling Permit «include» Review Permit Application
- Renew Selling Permit «include» Review Permit Application
- Browse Gun Catalog «include» View Gun Details
- Transfer Firearm Ownership «include» Update Firearm Sales Information

Extend «extend» Relationships

- Approve License Application «extend» Review License Application
- Reject License Application «extend» Review License Application
- Approve Permit «extend» Review Permit Application
- Reject Permit «extend» Review Permit Application
- Request Dealer Suggestion «extend» View Gun Details

1(b) Use Case Diagram



1(c) Use Case Descriptions

1) Register / Login

Basic:

Choose Register/Login → Enter email & password → System verifies credentials
→ User authenticated.

Alternate:

- Wrong password → Show error + reset option
- Duplicate email → Suggest Login

2) Apply for Firearm License

Basic:

Login → Open license application → Fill personal & firearm details → Upload documents → Submit application.

Alternate:

- Missing documents → Prompt user to upload
- Invalid data → Show validation error

3) Renew Firearm License

Basic:

Login → Select renew license → Verify existing license → Submit renewal request.

Alternate:

- License expired beyond renewal period → Show rejection message

4) Check License Status

Basic:

Login → Open license status → System retrieves application status → Display result.

Alternate:

- No application found → Show notification

5) Browse Gun Catalog

Basic:

Open catalog → System displays available firearms → User browses list.

Alternate:

- No firearms available → Show empty catalog message

6) View Gun Details

Basic:

Select firearm → System shows specifications, price, and availability.

Alternate:

- Firearm record unavailable → Show error message

7) Request Authorized Dealer Suggestion

Basic:

View gun details → Request dealer suggestion → System lists authorized dealers with stock.

Alternate:

- No dealer available → Show unavailability message

8) Apply for Selling Permit

Basic:

Dealer login → Open permit application → Enter business details → Submit request.

Alternate:

- Incomplete business info → Prompt correction

9) Renew Selling Permit

Basic:

Dealer login → Select permit renewal → Confirm details → Submit renewal.

Alternate:

- Permit expired beyond allowed period → Show renewal denial

10) Check Selling Permit Status

Basic:

Dealer login → View permit status → System displays current status.

Alternate:

- Permit not found → Show alert

11) Review License Application

Basic:

Authority login → Open pending applications → Review documents & details.

Alternate:

- Incomplete application → Mark for resubmission

12) Approve License Application

Basic:

Review application → Approve request → System updates license status.

Alternate:

- Legal issue detected → Redirect to rejection

13) Reject License Application

Basic:

Review application → Reject request → System records reason & notifies user.

Alternate:

- Missing rejection reason → Prompt authority to provide one

14) Review Selling Permit Application

Basic:

Authority login → Open permit requests → Verify dealer compliance.

Alternate:

- Missing documents → Request resubmission

15) Approve Selling Permit

Basic:

Review permit → Approve → System activates dealer permit.

Alternate:

- Compliance issue found → Redirect to rejection

16) Reject Selling Permit

Basic:

Review permit → Reject → System records reason & notifies dealer.

Alternate:

- Rejection reason missing → Prompt authority

17) Update Firearm Sales Information

Basic:

Dealer login → Enter sale details → Submit update → System saves record.

Alternate:

- Invalid firearm ID → Show error

18) Transfer Firearm Ownership

Basic:

Dealer login → Select firearm → Verify buyer license → Confirm transfer.

Alternate:

- Buyer license invalid → Block transfer

19) Search Firearms Related to Criminal Activity

Basic:

Authorized user login → Enter search criteria → System returns matching records.

Alternate:

- No match found → Show empty results

20) Manage Firearm Inventory

Basic:

Authorized user login → View inventory → Add/Edit/Remove firearm records.

Alternate:

- Unauthorized modification → Deny action

21) Monitor Welfare Usage

Basic:

Authority login → Access usage reports → Review compliance data.

Alternate:

- Data unavailable → Show system notice

22) Manage User Accounts

Basic:

Admin login → View users → Add/Update/Deactivate accounts.

Alternate:

- Invalid role assignment → Show error

23) Manage Firearm Database

Basic:

Admin login → Access firearm records → Add/Edit/Delete data.

Alternate:

- Duplicate firearm entry → Block insertion

24) Manage Dealer Records

Basic:

Admin login → Access dealer data → Update or verify records.

Alternate:

- Dealer record not found → Show error

25) Maintain System Database

Basic:

Admin login → Run maintenance tasks → Backup and optimize database.

Alternate:

- Maintenance failure → Log error & notify admin

2. Design Methodology Used

Design Methodology Used: SDLC (System Development Life Cycle)

For the Firearm Information Repository System (FIRS), the System Development Life Cycle (SDLC) methodology is used to guide the development process from planning to maintenance.

Why SDLC Is Suitable for This Project

SDLC is best suited for this project because FIRS is a government-regulated, data-sensitive system that requires clear documentation, strict validation, and controlled workflows.

Key Reasons

1. Clear and Well-Defined Requirements

Firearm licensing, permit approval, and ownership transfer follow strict legal rules. SDLC works best when requirements are clearly defined and must be followed accurately.

2. Structured and Sequential Approach

SDLC ensures each phase is completed and verified before moving to the next, reducing errors in critical modules like license approval and criminal firearm searches.

3. Strong Documentation

Government and regulatory systems require detailed documentation. SDLC emphasizes formal artifacts such as SRS, UML diagrams, test plans, and reports.

4. Risk Reduction

By validating requirements and design early, SDLC reduces risks related to security breaches, incorrect approvals, and data inconsistency.

5. Ease of Maintenance

The maintenance phase allows continuous updates such as new firearm laws, policy changes, and system enhancements.

SDLC Phases Applied to FIRS

1. Requirement Analysis

Identify stakeholders (Customers, Dealers, Government Authority, Admin)

Define functional and non-functional requirements

Analyze legal and compliance constraints

2. System Design

Create use case diagrams, class diagrams, and database design

Define system architecture and security controls

3. Implementation

Develop modules such as user management, licensing, dealer operations

Integrate database and authentication mechanisms

4. Testing

Unit testing of individual modules

Integration testing between system components

Validation of role-based access control

5. Deployment

Deploy the system in a controlled environment

Migrate and initialize firearm and user data

6. Maintenance

Fix bugs and security issues

Update system according to new firearm regulations

Why SDLC Over Agile for This Project

- Requirements are stable and regulation-driven
- Frequent requirement changes are not expected
- Formal approvals and documentation are required
- High importance of security, accuracy, and compliance

Conclusion

The SDLC methodology provides a systematic, disciplined, and secure development process for the Firearm Information Repository System, making it the most appropriate choice for a legally sensitive and structured system.

3. Project Scheduling – Gantt Chart

3(a) Project Breakdown Table

Phase	Detailed Task	Predecessor	Duration (Weeks)
Analysis	A Define system objectives	–	2
	B Study of laws & regulations	–	2
	C Requirement analysis & SRS	A	2
	D Feasibility & compliance study	B	2
Development	E System & database design	C	3
	F User authentication module	E	2
	G License management & inventory module	F	2
	H Dealer transaction module	E	2
	I Search & reporting module	H	2
Implementation	J System integration	G	2
	K Testing & validation	J	1
	L Deployment & maintenance	K	1

3(b) Gantt Chart (Textual View)

